WHEN, WHY, AND HOW FAR IT HELPS

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through intermediate emotional responses

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USER A	2.0 / 7.0	Sad / Distasteful
USER B	5.0 / 7.0	Intellectual / Motivated
USER C	6.0 / 7.0	Impressed / Nostalgic

02 — AGENDA

What we'll cover today.

03 — INTRODUCTION

From average taste to personal taste.

GENERIC MODEL
Image Aesthetic Assessment

IAA Landscape

3.5 / 7.0

IAA: one score, the average opinion

PERSONALIZED MODEL
Personal Taste (PIAA)

PIAA Landscape

User A	User B	User C
2.0/7	5.0/7	6.0/7

PIAA: how much does THIS person like it?

THE CHALLENGE
Black-Box Paradigm

User ID
↓
Black-B? model
Predicted Score

accurate, but cannot explain WHY

Applications: Photo Recommendation • Album Curation • Ranking AI-Generated Images

04 — OUR APPROACH

We define preference through feeling.

Input target image
Image

→

7
Emotions
nostalgic, distasteful...

→

Linear Formula
$y = w_0 + \sum_{e}^{7} w_e x_e$
readable per-person weights

→

Preference Score
(Aesthetic Rating)

"This image makes me feel X, so I like it this much."

05 — PIPELINE

Two pathways: Direct and Hybrid.

Direct Pathway: photo → score

Hybrid Pathway: photo → 7 emotions → score

Image
INPUT
6,526 photos
art / fashion / landscape

→

Backbone
DIRECT
image → features
Qwen-4B VLM features

→

SHARED (GENERIC)
7 Emotions
HYBRID
how does this photo feel?
impressed · intellectual · amused · nostalgic · sad · distasteful
trained on 104 general users

shared (trained on general users)

→

PERSONALIZED (PER-USER)
Personal Formula
MEDIATOR
7 weights, one per emotion
trained on target's training split

this is what we can read

→

Score
OUTPUT
0 – 6 score
predicted user aesthetic score

The gap between Direct and Hybrid = what emotion adds

06 — DATASET & PROTOCOL

One dataset deep enough per person.

129 **People** (Evaluators)

6,526 **Images** (Unique Photos)

3 **Categories** (Art, Fashion, Landscape)

1. **Deep Per Person**
   Each evaluator rated hundreds of images → we can fit a robust, personal formula.

2. **Test-Retest Design**
   A subset of images were rated twice by the same user → used later to calculate the ceiling.

leak-free: test users & images never seen in training

SECTION DIVIDER

Part 2 — Does it actually work?

08 — FINDING 1

Going through emotion helps 93% of users.

Direct CCC: 0.293 → **Hybrid CCC: 0.359**

+0.066 **Gain (Delta CCC)**

helps 92.8% (93%) of people

Dose-Response Behavior:
As the number of ratings per user increases, the accuracy improvement becomes larger and more stable. Emotion mediation prevents model overfitting when adapting to small amounts of personal taste data.

Dose-Response Support Curve

Delta (Hybrid - Direct) increases with user ratings size

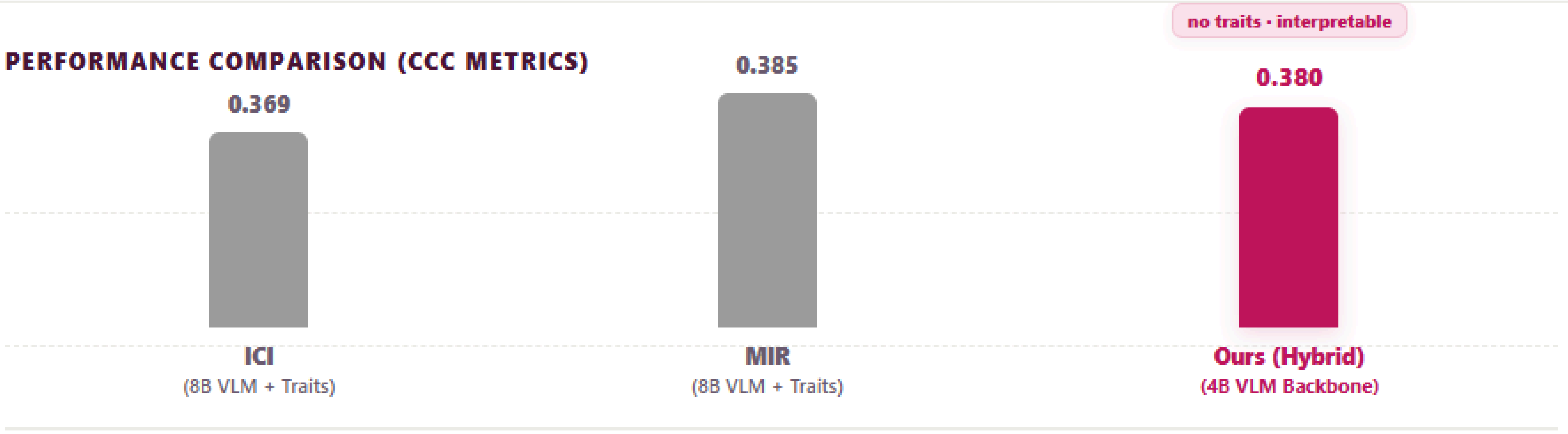
Direct Baseline: 0.293 | Hybrid Pathway: **0.359** | Accuracy Gain: **+0.066 (helps 93%)**

more personal data → more help

09 — FINDING 3

We match trait methods, without traits.

PERFORMANCE COMPARISON (CCC METRICS)



Core Novelty & Contribution

We achieve competitive predictive accuracy matching SOTA 8B baselines without requiring intrusive, multi-question user personality surveys (like the Big Five index).

STRONG ARGUMENT FOR USER PRIVACY

Protecting User Privacy

Predicting taste through intermediate aesthetic emotions allows us to completely bypass the storage or assessment of sensitive user profile traits.

- Stronger backbone ≠ always better (Qwen2 ≈ Qwen3)
- Fine-tuning not needed — matches Ryu & Yanaka

It works. Now three questions.

1. When does it help?
2. How far can it go?
3. Is the gain real?

11 — FINDING 4

It helps more when we read emotions well.

CORE DRIVING FACTOR

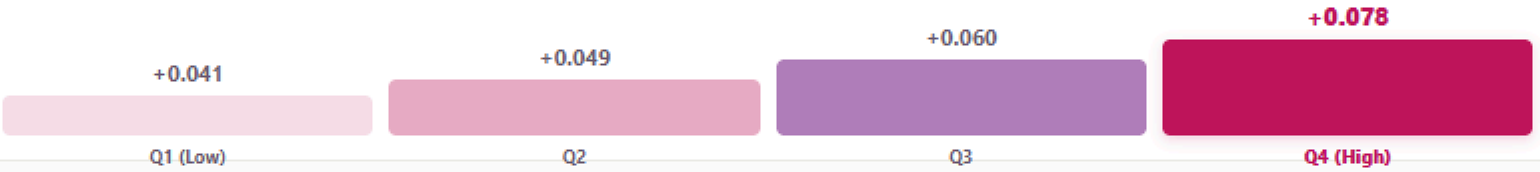
One clean factor: emotion accuracy

The more accurately we predict emotions, the more the emotion pathway helps.

Spearman r
+0.273

EMOTION PREDICTABILITY (EMO_r) QUARTILE GAIN

gain rises **0.041 → 0.078** across groups



Gain vs Emotion Predictability (emo_r)

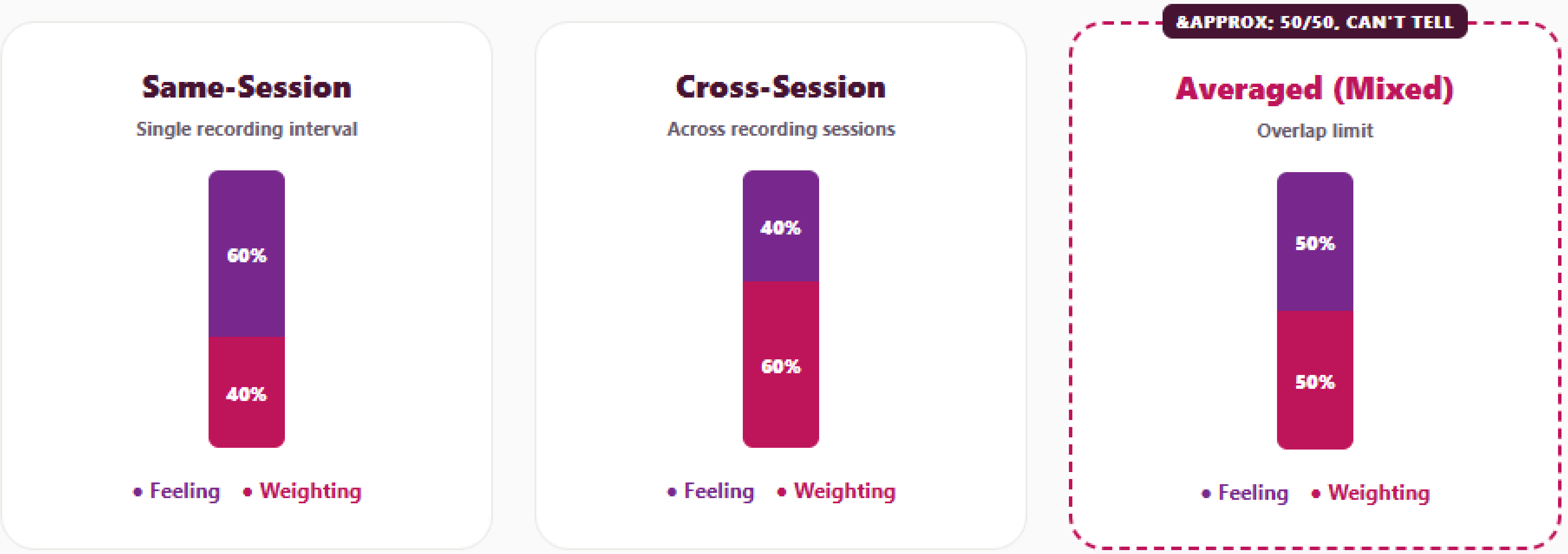
Delta (Hybrid - Direct) scales with emotion prediction accuracy (Spearman +0.273)

Spearman r: **+0.273** | Q1 Gain: +0.041 | Q4 Gain: **+0.078**

the rival explanation was a confound

12 — FINDING 2

Feeling vs weighting: it depends how we measure.



so we report it honestly, as a supporting result with ranges

13 — FINDING 6

The realistic ceiling is 0.64.

0.639 **Deployment-Realistic Ceiling**
Aesthetic predictability ceiling measured across distinct sessions (Cross-Session).

PERFORMANCE RATIO

59% **we reach 59% of it (0.380 / 0.639)**

averaging emotions raises it — the limit is measurement noise

Human Noise Ceiling Estimates

Comparing different noise ceiling definitions with Hybrid model performance

- Same-Session (Theoretical): Higher (~0.81)
- Cross-Session (Deployment-Realistic): 0.639
- Our Hybrid Model: 0.380 (59%)

14 — FINDING 8

Emotion makes personalizing worth it.

THE 50-RATINGS THRESHOLD

When is Personalization Worthwhile?

Below approximately 50 user ratings, individual personalization performs worse than the population average model due to data sparsity.

CRITICAL PRACTICAL INSIGHT

"Unless you use emotion mediation (Hybrid), personalizing with few ratings (under 50) is worse than simply using the population average."

Cold-Start Threshold Curve

Hybrid personalization begins to strictly outperform population baseline after 50 user ratings

Population baseline: **Dashed Pink** | Hybrid Pathway: **Beats population at 50** | Direct Baseline: **Sub-population**

Direct never beats the crowd. Hybrid does, from 50 ratings.

15 — VALIDITY CHECK

Validation: placebo control is random noise.

CONTROL EXPERIMENT RESULTS (CCC)

Ours (Real Emotions)	**0.380**
Placebo (Random Noise / PCA)	0.160

PLACEBO PCA PROJECTION CHECK:

$$f_{PCA}(i) = f_{VLM}(i) \cdot W_{PCA}$$

Placebo Controls Performance

Personalization gains drop when real emotion data is replaced by shuffles or noise

Ours (Real): 0.380 | Placebo (Noise): 0.160 | Difference (Gain Gap): +0.220

Ours: 0.380, Placebo: 0.160. Emotion semantics are real.

16 — CONCLUSION

Key takeaways.

1 Explainable PIAA
Provides readable weights per person, achieving positive performance gains for 93% of users.

2 No Intrusive Traits
Matches SOTA baselines using emotional intermediate representations, fully protecting user privacy.

3 Realistic Ceiling
Temporal ceiling is capped at 0.64 due to measurement noise rather than model capacity.

4 Cold-Start Limit
Requires approximately 50 ratings to outperform the generic population average model.

"Emotion is a powerful, explainable bridge to subjective taste."

17 — OUTRO

Future work.

- **Context-Dynamic Weighting**
  Weights dynamically update depending on context, ambient conditions, or user mood shifts over time.

- **Active Learning Querying**
  Query users strategically to fit weights faster, reducing the 50-ratings cold-start threshold significantly.

Thank you

QUESTIONS & ANSWERS

Decomposition detail.

Q&A-A **VARIANCE DECOMPOSITION COMPARISON**

Definition	Perception %	Weighting %
Same-Session	60% [55%, 65%]	40% [35%, 45%]
Cross-Session	40% [34%, 46%]	60% [54%, 66%]
Averaged (Mixed)	**50% [42%, 58%]**	**50% [42%, 58%]**

3 Predictor Categories:

- **Perception (Feeling differences):** How different users perceive the emotional content of an image.
- **Weighting (Formula differences):** How differently users weigh emotions when forming aesthetic preferences ($w_{u,e}$).
- **Intercept (Baseline rating differences):** Differences in user baseline rating scales (w_0).

PERSONALIZED PIAA REGRESSION MODEL:

$$Score(u,i) = w_0 + \sum_{e=1}^{7} w_{u,e} f_e(i)$$

Noise Ceiling Estimates.

Q&A-B **NOISE CEILING CONFIGURATIONS**

Configuration Case	Ceiling (CCC)
Case A: Same-Session, Single Annotation	0.81
Case B: Cross-Session, Single Annotation (Realistic)	**0.64**
Case C: Same-Session, Multi-Annotation Averaged	0.86
Case D: Cross-Session, Multi-Annotation Averaged	0.69

Statistical Rigor:

Per-User Ceiling: Median self-agreement is 0.693 (CCC) when same users rate same images across sessions.

Ranking Stability: SROCC drops only marginally from 0.82 to 0.79, indicating ranking survives session noise well.

Kolmogorov-Smirnov Test: $p > 0.05$, confirming that rating distributions across different sessions remain stationary.

Leak-free validation.

Q&A-C **10-GROUP 5-FOLD CROSS-VALIDATION**

1. Disjoint User Splits:
129 users are split into 10 groups. One group is entirely held out for evaluation.

2. Stage 1 (Shared Emotion Model):
Trained on 9 groups. The held-out group's users and their rated images are never seen.

3. Stage 2 (Personal Fitting):
Evaluated on the held-out group using 5-fold CV to adapt weights.

Disjoint Group Splits:

Adaptation split: 100 images per user used to customize the emotion weights.

Validation split: 10 images per user used for validation.

Evaluation split: 19 images per user used to report final model accuracy.

Partial correlation.

Q&A-D **PARTIAL CORRELATION ANALYSIS**

Metric	Raw Correlation	Partial Correlation
Pearson r	0.42 ($p < 0.01$)	**0.38** ($p < 0.01$)
Spearman rho	0.39 ($p < 0.01$)	**0.35** ($p < 0.01$)

Why Partial Correlation?

We must control for the baseline model's performance. If we do not control for it, the correlation between emo_r (emotion predictability) and performance gain could be a statistical artifact (e.g. simply predicting worse users better).

The significant partial correlation confirms that **emotion accuracy is the primary driver of accuracy gain**, independent of the baseline's prediction room.

Design choices.

Q&A-E WHY LINEAR MODEL?

- **Cognitive Science basis:** Aesthetic judgment behaves linearly when mediated by emotions (Iigaya et al., Nature Communications 2020).
- **Explainability:** Yields a readable per-person weight vector (w_{ue}) mapping specific emotions directly to taste.
- **Regularization:** Ridge regression handles collinearity between the 7 emotions effectively. Non-linear models (MLPs) did not show statistically significant CCC improvements.

Q&A-E WHY EXACTLY 7 EMOTIONS?

- **Taxonomy:** Selected from the 21 emotional sub-scales in the AESTHEMOS taxonomy.
- **Circular Reasoning Avoidance:** Adjectives like "beautiful" or "like" are intentionally excluded to prevent predicting preference from preference.
- **Coverage:** Spans positive (motivated, amused), negative (sad, distasteful), and intellectual aesthetic emotions.

Related work.

Q&A-F **COMPARISON WITH PRIOR PIAA WORKS**

Prior Work	Approach	Our Key Difference
Ryu & Yanaka (2022)	Uses Big Five personality traits to group users.	**Traits-free:** We bypass sensitive personality tests, using visual emotion semantics.
Liu & Wagemans (2020)	Focuses on shared (generic) consensus emotional ratings.	**Personal weights:** We adapt the weighting profiles *per user* for aesthetics.
Lan et al. (2021)	Deep end-to-end learning to predict personal score.	**White-box explainability:** Our formula yields readable, interpretable user preferences.
Iigaya et al. (2020)	Linear weighting on low-level features (brightness, contrast).	**High-level VLM features:** We model semantic visual emotions using pre-trained VLMs.